

1. Quiz (15 Questions – Mix of MCQ, True/False, Short Answer)

Concept Check – Java Object & Class

Part A: Multiple Choice (5 questions)

1. What does the **new** keyword do in Java?
 - A. Declares a new class
 - B. Creates a new method
 - ☒ C. Creates a new object from a class
 - D. Deletes an object
2. Which of the following is a **valid Java constructor** for a class named **Dog**?
 - A. `public void Dog()`
 - ☒ B. `public Dog()`
 - C. `void Dog()`
 - D. `constructor Dog()`
3. What is the default return type of a constructor?
 - A. `int`
 - B. `void`
 - C. `String`
 - ☒ D. No return type
4. What is the output of **the below code** ?

```
class Constructor
{
    static int x = 1;
    public Constructor()
    {
        x = 5;
    }

    public void showX()
    {
```

```

    System.out.println("x : " + x);
}

public static void main(String[] args)
{
    Constructor obj = new Constructor();
    System.out.println(x);
    obj.showX();
}
}

```

constructor is called, x=5

A. 5

x : 5

B. x : 5

x : 5

C. Compilation Error

D. Run time error

5. Which of these is **NOT** a benefit of OOP?

A. Code reuse ✓

B. Scalability ✓

C. Better security

D. Slower performance

Part B: True/False (5 questions)

6. A Java class can have multiple constructors. **True** (constructor overloading)

7. Objects in Java are created from classes. **True**

8. You must always define a **main()** method inside a class to compile it. **False**,
 → run

9. Instance variables are declared inside methods. **False**
 local

10. The this keyword refers to the current object. *True*

Part C: Short Answer (5 questions)

11. Define what an object is in Java. *-instance of a class*

12. What is the difference between a class and an object?

- A class is a template that defines properties and behaviors.
- An object is an instance of a class
 - holds actual values

13. Why do we use constructors in a class? *- initialize objects when they are created*
set initial values for instance variables and perform setup procedures

14. Write the code to create an object of **Car** class.

Car myCar = new Car();

15. Name two things that a class can contain.

- Variables
 - methods
-



2. Code Examples

Example: Student Class

Unset

```
public class Student {
    String name;
    int age;

    public Student(String studentName, int studentAge) {
        name = studentName;
        age = studentAge;
    }

    public void printDetails() {
        System.out.println("Name: " + name + ", Age: " + age);
    }

    public static void main(String[] args) {
```

```

        Student s1 = new Student("Alya", 20);
        s1.printDetails();

        Student s2 = new Student("Zafran", 22);
        s2.printDetails();
    }
}

```

Challenge Task: Book Class (Student Practice-Fix the errors)

Unset

```

public class Book {
    String title;
    String author;
    int yearPublished;

    public Book(int y, String a, String t) {
        title = t;
        author = a;
        yearPublished = y;
    }

    public void displayBookInfo() {
        System.out.println("Title: " + title + ", Author: " +
author + ", Year: " + yearPublished);
    }

    public void isClassic() {
        return yearPublished < 2000;
    }

    public static void Main(String[] args) {
        Book 1b = new Book("To Kill a Mockingbird", "Harper Lee", 1960);
        1b.displayBookInfo();
    }
}

```

Book 1b = new Book (1960, "Harper Lee",
"To Kill a Mockingbird")

```
        System.out.println("Classic? " + isClassic);
    }
}
```

Check and Fix the errors in Book.java . The output should be as below :

Output:

```
Title: To Kill a Mockingbird, Author: Harper Lee, Year: 1960
Classic? true
```



3. Student Worksheet

Title: Object Class in Java - Practice Worksheet

Section A – Fill in the Blanks

1. A class is a blueprint for creating objects.
2. An object is an instance of a class.
3. The new keyword is used to create objects.
4. A constructor has the same name as the class and no return type.
5. Use the `this` keyword to refer to the reference object.

Section B – Write Code

Create a `Movie` class with:

- Fields: `title`, `genre`, `rating` (double)
- Constructor
- Method: `printMovieInfo()`

done ✓

Then create 2 `Movie` objects in `main()` and call the method.

Section C – Debug the Code

Find and fix the errors:

Unset

```
public class Car {
    String model;
    int year;

    public void Car(String m, int y) {
        model = m;
        year = y;
    }

    public void printInfo() {
        System.out.println("Model: " + model + ", Year: " +
year);
    }

    public static void main(String[] args) {
        Car myCar = new Car("Honda", 2022);
        myCar.printInfo();
    }
}
```

Section D – Reflection (Short Answer)

1. What was easy about today's coding tasks? *Fix errors*
 2. What was hard or confusing? *Writing definition is a little bit hard, since I don't know*
 3. Rate your confidence (1–10) with creating Java objects. *10* *how to explain.*
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